

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2412

Cedar Tap 138 kV
August 19, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2412. The Interconnection Customer proposes a 250.0 MW solar photovoltaic facility interconnecting at the Cedar Tap 138 kV substation in the Kentucky area. The request was received on May 29, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 432 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,067,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: LOOKOUT POINT-LANK 138 kV
Contingency: SPRING CRK RD-ATOKA 138 kV Line
Loading: 110.5% (Rating: 227 MVA)
DFAX: 6.7%
Pre-project loading: 103.1%
2. Facility: TURKEY CRK-TOM WATKINS 138 kV Line
Contingency: HARRIS-SPRING CREEK RD 138 kV
Loading: 110.4% (Rating: 388 MVA)
DFAX: 7.4%
Pre-project loading: 105.6%
3. Facility: MEEKER-TURKEY CREEK 138 kV
Contingency: SPRING CREEK RD-HARRIS 138 kV line
Loading: 110.2% (Rating: 396 MVA)
DFAX: 7.4%
Pre-project loading: 105.5%
4. Facility: MEEKER-SIDNEY COAL 138 kV Line
Contingency: SPRING CREEK RD-HARRIS 138 kV L
Loading: 109.7% (Rating: 413 MVA)
DFAX: 7.4%
Pre-project loading: 105.2%
5. Facility: TULE-GRANITE CITY WORKS 138 kV L
Contingency: WISSOTA-SAN ANDREAS SUBSTATION 138 kV
Loading: 102.5% (Rating: 276 MVA)
DFAX: 5.6%
Pre-project loading: 97.4%

6. Facility: BRAWLEY-LOOKOUT PT 138 kV
 Contingency: ATOKA-SPRING CREEK RD 138 kV
 Loading: 101.7% (Rating: 231 MVA)
 DFAX: 6.7%
 Pre-project loading: 94.5%
7. Facility: SPRING CREEK ROAD-ATOKA 138 kV L
 Contingency: MEEKER-SIDNEY COAL 138 kV
 Loading: 101.0% (Rating: 433 MVA)
 DFAX: 10.4%
 Pre-project loading: 95.0%
8. Facility: SANTA ROSA BEACH-TULE 138 kV
 Contingency: SAN ANDREAS-WISSOTA 138 kV
 Loading: 101.0% (Rating: 291 MVA)
 DFAX: 5.6%
 Pre-project loading: 96.2%
9. Facility: SANTA ROSA BEACH-W MOUNT VERNON 138 kV
 Contingency: WSSOTA-SAN ANDREAS 138 kV line
 Loading: 100.8% (Rating: 308 MVA)
 DFAX: 5.6%
 Pre-project loading: 96.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOOKOUT POINT-LANK 138 kV	Rebuild line with higher-capacity conductor	\$10,767,000	9% (\$947,000)
TURKEY CRK-TOM WATKINS 138 kV Line	Rebuild line with higher-capacity conductor	\$8,806,000	8% (\$729,000)
MEEKER-TURKEY CREEK 138 kV	Rebuild line with higher-capacity conductor	\$644,000	8% (\$53,000)
MEEKER-SIDNEY COAL 138 kV	Reconductor line	\$2,251,000	8% (\$186,000)
TULE-GRANITE CITY WORKS 138 kV	Reconductor line	\$9,169,000	5% (\$479,000)
BRAWLEY-LOOKOUT PT 138 kV	Reconductor line	\$1,675,000	9% (\$147,000)
SPRING CREEK ROAD-ATOKA 138 kV L	Reconductor line	\$4,777,000	8% (\$394,000)
SANTA ROSA BEACH-TULE 138 kV	Reconductor line	\$1,651,000	5% (\$86,000)
W MOUNT VERNON-SANTA ROSA BEACH 138 kV	Reconductor line	\$885,000	5% (\$46,000)
	Total allocated to project		\$3,067,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.